CN ASSIGNMENT

Tutorial sheet - 03

DATE: 05/11/23
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• Section - A (5*2=10) (22BCON1068)

Q1. What do you mean by network layer?

Ans The network layer is the third layer of the open system interconnection (OSI) model of computer networking. it is responsible for routing packets from the source to the destination host across multiple networks.

The network layer provides the following services:

- Logical addressing
- Routing
- Fragmentation and reassembly
- Error checking and correction.

Q2. What is the Routing algorithm?

Ans A Routing algorithm is a procedure that determines the best path for packets to take from the source to the destination host in a computer network. Routing algorithms are used by routers to make decisions about which outgoing link to send a packet on.

• The Routing algorithms can be classified as follows.

(i) Adaptive Algorithms
(ii) Non-adaptive algorithms.

Q3. What is Flooding?

Ans Flooding is a non-adaptive routing technique following this simple method: When a data packet arrives at a router, it is sent to all the outgoing links except the one it has arrived on.

Flooding is a simple and effective way to ensure that all nodes in a network receive a packet, regardless of their location, However, it can also be inefficient, as it can lead to duplicate packets being sent to the same nodes.

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- Flooding is used in a variety of networking applications including:
  - (i) Routing updates
  - (ii) Multicasting
  - (iii) Bridging

Q4. Write any three names of Routing Algorithms.

Ans. Here are three names of routing algorithms:

(i) Distance-vector routing algorithms:- RIP (Routing information Protocol) and EIGRP (Enhanced Interior Gateway Routing Protocol) are examples of distance-vector routing algorithm.

(ii) Link-state routing algorithms:- OSPF (Open Shortest Path First) and IS-IS (Intermediate system-to-Intermediate system) are examples of link-state routing algorithm.

(iii) Path-vector routing protocols:- BGP (Border-Gateway Protocol) is an example of a path routing protocols.

Q5. What is DHCP?

Ans. DHCP stands for Dynamic Host Configuration Protocol. it is a network protocol that automatically assigns IP addresses and other network configuration parameters to devices on a network.

DHCP is a client-server protocol, which means that there is a DHCP server that manages the pool of available IP addresses and a DHCP client on each device on the network. When a device first connects to a network, it sends a DHCP request to the DHCP server. The DHCP server then responds with an IP address and other network configuration parameters, such as the subnet mask and default gateway.

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DHCP can be implemented on local networks as well as large enterprise network. DHCP is the default Protocol used by the most routers and networking equipment. DHCP is also called RFC (Request for comments 2131.

- Section- B(3*7=21)

Q1. Explain the shortest path routing algorithm.

Ans. The shortest path routing algorithm is a technique used to find the shortest path between two nodes in a network. it is used ~~to~~ by routers to determine the best path to forward packets.

The shortest path routing algorithm works by maintaining a table of distances between each node in the network. The table is updated as the router learns about the topology of the network. When a packet arrives at the router, the router uses the distance table to determine the best path to forward the packet.

in this algorithm, to select a route, the algorithm discovers the shortest path between two nodes. it can use multiple hops, the geographical area in kilometers or labelling of arcs for measuring path length.

The labelling of arcs can be done with mean queuing, transmission delay for a standard test packet on an hourly basis or computed as a function of bandwidth, average distance traffic, communication cost, mean queue length, measured delay or some other factors.

in shortest path routing, the topology communication network is defined using a directed weighed graph. The nodes in the graph define

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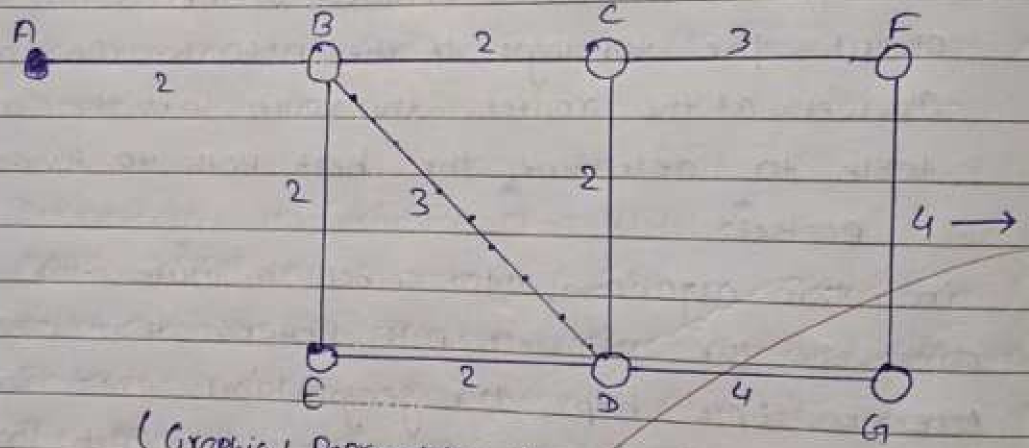
switching components and directed arcs in the graph define communication connection betn. switching components. Each arce has a weight that defines the cost of sharing a packet betn two nodes in a specific direction.

This cost is usually a positive value that can denote such factors as delay, throughput, error rate, financial costs, etc. A path betn. two nodes can go through various intermediary nodes and arcs.

For example: Dijikstra uses the nodes labelling with its distance from the source node along the better-known route. The labelling graph is displayed in the figure.



(Graphical Representation of nodes with labeled path)

→ it can be done in various passes as follows, with A as the source

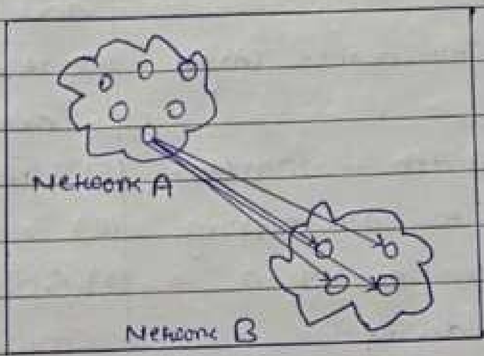
- Pass 1. B(2,A), C(∞,-), F(∞,-), e(∞,-), d(∞,-), G60
- Pass 2 .. B(2,A), C(4,B), D(5,B), E(4,B), F(∞,-), G(∞,-)
- Pass 3. B(2,A), C(4,B), D(5,B), E(4,B), F(7,C), G(9,D)

We can see that there can be two paths between A and G. One follows through ABCFG and the other through ABDG. The first one has a path length of 11, while the second one has 9.

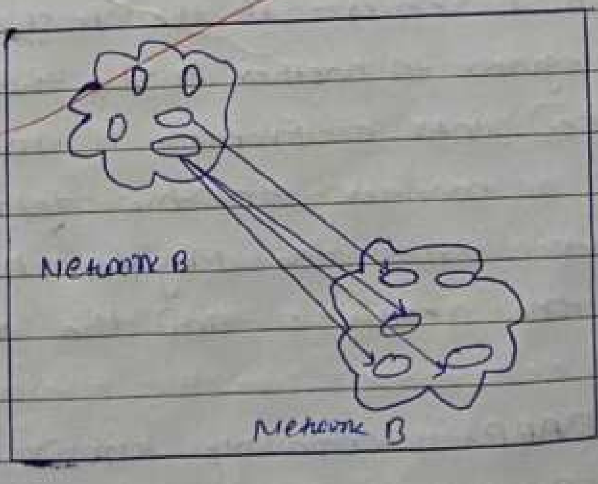
Q2. What is the difference betn broadcast and multicast routing algorithms ?

Ans. • Broadcast : Broadcast transfer (one-to-all) techniques and can be classified in two types : limited Broadcasting, Direct Broadcasting. in broadcasting mode, transmission happens from one host to all other hosts connected on the LAN. The devices such as bridge uses this. The protocol like ARP implement this. Broadcast routing algorithm sends a packet ~~on the network~~. to all devices on the network.

~~Multicasting~~ :



• Multicasting : Multicasting R. A. send a packet to a specific group of devices on the network. this is useful for applications such as video streamming and file sharing. multicasting routing algorithm are more efficient than broadcast routing algorithm, as they only send packets to the devices that have requested them.



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- Differena betn. Broadcast and Multicast :

Broadcast	Multicast
(1) it has one sender and Multiple receiver.	(1) it has one or more senders and multiple receiver.
(2) it sent data from one device to all the other devices in a network.	(2) it sent data from one device to multiple devices.
(3) it works on star and bus topology.	(3) it works on star, mesh, tree and hybrid topology.
(4) it scale well across large networks.	(4) it does not scale well across large network.
(5) its bandwidth is wasted.	(5) it utilizes bandwidth efficiently
(6) it has one-to-all mapping	(6) it has one-to-many mapping.
(7) Hub is an example of a broadcast device.	(7) Switch is an example of a multicast device.

Q2. Differentiate betn. ARP and RARP.

Ans. ARP: ARP is commonly "Address Resolution Protocol". it is a form of network Layer Protocol. As the ARP Protocol is a dynamic mapping Protocol, every host on the network is aware of another logical address of the host server. if the host requires transfering an IP datagram to another host, the IP datagram must be enclosed into a Frame. it must be accomplished in such a manner that the datagram may easily flow ~~across~~ across the Physical network that exists between the receiver and the transmitter.

- RARP: RARP is commonly known as a "Reverse Address Resolution Protocol". The RARP Protocol is also a type of network layer protocol. A TCP/IP protocol enables any host to get its true IP address from the network server. As the names defines, the RARP protocol is an extended version of the ARP. it is usually the opposite of the ARP protocol.

• Differentiate :

	ARP	RARP
(1)	A Protocol used to map an IP address to a physical (MAC) address.	A protocol used to map a physical (MAC) address to an IP address.
(2)	To obtain the MAC address of a network device when only its IP address is known.	To obtain the IP address of a network device when only its MAC address is known.
(3)	client broadcasts its IP address and requests a MAC address, and the server responds with the corresponding MAC address.	(3) client broadcasts its MAC address and requests an IP address and the server responds with the corresponding IP address.
(4)	IP address	(4) MAC address
(5)	widely used in modern networks to resolve IP addresses to MAC addresses.	(5) Rarely used in modern networks as most devices have a pre-assigned IP address.

• Section – C (3x11 = 33)

(1) Explain NAT in detail with diagram.

Ans. Network Address Translation (NAT) is a technique for translating one ore more private IP addresses into one or more public IP addresses. This allow multiple devices on a private network to share a single public IP address when accessing the internet.

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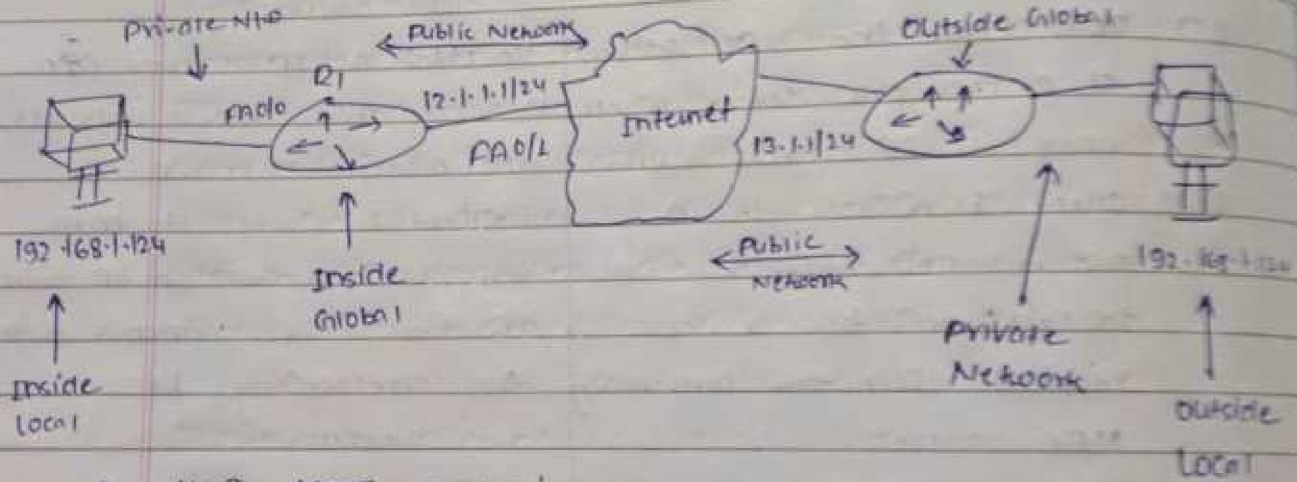
NAT is typically implemented on a router or firewall
that is located at the edge of the private network
when a device on the private network wants to send
a packet to the internet, the NAT router translates
the device's private IP address to the router's
Public IP address. The NAT router also keeps
track of which internal devices initiated the outgoing
connections so that it can route incoming packets
back to the correct devices.



- How NAT works:

(1) A device on the private network wants to send a packet to the internet.

(2) The packet is sent to the NAT Router.

(3) The NAT Router translates the device's Private IP address to the router's Public IP address.

(4) The NAT router also adds a port number to the packet

(5) The NAT router forwards the packet to the internet.

(6) When a packet is returned from the internet, the NAT router looks up the port number in its NAT table to determine which internal device initiated the connection.

(7) The NAT router forwards the packet to the correct internal device.

- Types of NAT : There are three types of NAT.

(1) Static NAT : in this, a single unregistered (Private) IP address is mapped with a legally registered (public) IP address i.e; one-to-one mapping between local and global addresses. This is generally used for web hosting. These are not used in organizations as there are many devices that will need internet access and to provide internet access, a public IP address is needed.

suppose, if there are 3000 devices that need access to the internet, the organization has to buy 3000 public addresses that will be very costly.

(2) Dynamic NAT : in this type of NAT, an unregistered IP address is translated into a registered (public) IP address from a pool of public IP address. if the IP address of the pool is not free, then the packet will be dropped as only a fixed number of private IP addresses can be translated to public addresses.

(3) Port Address Translation (PAT) : This is also known as NAT overload. in this, many local (private) IP address can be translated to a single registered IP address. Port numbers are used to distinguish the traffic i.e; which traffic belongs to which IP address. This is most frequently used as it is cost-effective as thousands of users can be connected to the internet by using only one real global (public) IP address.

Q2. Describe internet protocols (IPv4, IPv6) in detail.

Ans. Internet Protocol (IP) is the essential protocol for routing and addressing of data packets across networks. it serves as the foundation of the internet, enabling communication between devices worldwide. Two primary IP version are widely used: IPv4 and IPv6.

- IPv4 (Internet Protocol Version 4).
- overview :- IPv4 is the fourth revision of the internet Protocol and has been the backbone of the internet since its introduction in the 1980s. it utilize 32-bit addresses, which provide a total of approximately 4.3 billion unique addresses. This may seem like a vast number, but with the exponential growth of connected devices, IPv4's address space is quickly becoming depleted.

- Addressing Scheme :
  IPv4 addresses are represented as a sequence of four decimal number, each ranging from 0 to 255, seperated by periods. for instance, 192.168.1.100 is a valid IPv4 address. These numbers represent specific network segments and hosts within those segments.
- Header structure :
  IPv4 Packets contain a header that provides essential information for routing and data delivery. The header includes fields like source and destination IP addresses, packet type, time-to-live (TTL) and Payload length.

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- IPv6 (internet Protocol version 6):

IPv6, introduced in the late 1990s, is the successor to IPv4 and addresses the limitations of its predecessor. it employs 128-bit addresses, resulting in a virtually limitless address space, accomodating the ever-expanding internet.

- Addressing scheme: IPv6 addresses are represented as eight group of four hexadecimal digits, separated by colons. for example, 2001:0db8:85a3:0000:0000:8a2e:0370:7334 is an IPv6 address. These hexadecimal groups represent different parts of the address, including the network prefix, subnet, identifier, and interface identifier.
- Header structure: IPv6 packets have a simplified header compared to IPv4, reducing overhead and improving efficiency. The header includes fields like source and destination IP addresses, payload length and next header.
- Comparision of IPv4 and IPv6.

Feature	IPv4	IPv6
Address space	4.3 billion	virtually limitless
Address Format	32-bit decimal	128-bit hexadecimal
Security	Optional	Mandatory
Header size	20 bytes	40 bytes
Routing	Hierarchical	Flatter, more efficient
Deployment	widespread	Increasing adoption

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Q3. Write a short note on:

• ICMP protocol : The internet Control Message Protocol (ICMP) is a network layer protocol used by network devices, including routers, to send error Message and operational information. it is an essential component of the internet protocol suite, providing feedback about data transmission and network status. ICMP is used for reporting errors and management queries. it is a supporting protocol and is used by network devices like routers for sending error message and operations information. For example, the requested service is not available or a host or router could not be reached.

ICMP is used for error reporting if two devices connect over the internet and some error occurs, so. the router sends an ICMP error message to the source informing about the error. For example, whenever a device sends any message which is large enough for the receiver, in that case, the receiver will drop the message and replay back ICMP Message to the source.

• Link State Routing : it is a technique in which each router shares the knowledge of its neighbourhood with every other router in the inter network.

Three keys to understand link state routing algorithm:

(i) knowledge about the neighbourhood : instead of sending its routing table, a router sends the information about its neighbourhood only.

(ii) Flooding : Each router sends the information to every others router on the internetwork except its neighbours. This process is known as flooding.

(iii) information sharing : A router sends the information to every other router only when the change occurs in the information.

Link state routing is a dynamic routing algorithm in which each router shares its knowledge of its neighbors with every other router in the network. This information is used to construct a map of the network topology, which is then used to calculate the shortest path to each destination. Link state routing is a very responsive and efficient routing protocol, and it is widely used in large networks.

- Advantages of Link-State Routing:
  - → Loop - Free
  - → Fast convergence
  - → Scalable

- Disadvantages of Link-State Routing:
  - → High CPU and Memory Requirements.
  - → increased Network Traffic.

- examples of Link-State Routing protocols :—
  - → open shortest path first
  - → intermediate system to intermediate system (IS-IS).

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ASSIGNMENT - 03 (CN)

Tutorial Sheet - 04

• Sec-A (5*2 = 10)

Q1. What is OSPF routing?

Ans. OSPF stands for open shortest path first, it is a routing protocol that helps routers find the best path for sending data within a network. it's like a GPS for data packets, ensuring they reach their destination efficiently.

Q2. Representation of IPv6.

Ans An IPv6 address is 128 bits in length and is written as eight groups of four hexadecimal digits. Each group is seperated from the others by colons (:) as shown in figure.

2001:0df8:$\boxed{23F2}$:0000:0000:66ee:1336:1944

2	3	F	2

→ Every hexadecimal digit is equivalent to 4 bits.

each group of 4 hex digits represents 16 bits.

Q3. What do you mean by BOOTP?

Ans BOOTP stands for Bootstrap protocol. it is a network protocol used by network devices, such as computers and printers, to obtain their IP address and other configuration information during the boot process. BOOTP is designed to work in a network environment where a device may not have a permanent (static) IP address assigned.

Q4. What is path vector Routing?

Ans. A path vector routing protocol is a type of routing protocol that uses information about the paths to other networks to make routing decisions. This information, known as a path vector, includes the list of autonomous system (AS) that a packet must pass through to reach its destination. Path vector routing protocols use this information to determine the best path to a destination, taking into account factors such as the number of ASes, the weight of the links betn ASes, and the administrative policies of the ASes.

Q5. Define the term of internet working?

Ans internet working is the process or techniques of connecting different networks by using intermediary devices such as routers or gateway devices.

internetworking ensures data comm. among networks owned and operated by different entities using a common data comm. and the internet Routing protocol.

• Section - B (3×7 = 21)

Q1. Differentiate betn Shortest path routing and path vector routing.

Ans Shortest path Routing :-

• Concept :-

→ Each router maintains a complete picture of the network topology.

→ This topology information includes the cost of each link betn routers.

• Characteristics :-

→ Fast convergence

→ Scalable

→ Loop free

→ High bandwidth.

- Example :-
  → Open shortest path first (OSPF)
  → (IS-IS)

- Path vector Routing :-
- concept :-
  → Each router only knows about its immediate neighbors and the paths to other networks.
  → Routers advertise their paths to other routers using routing updates.
- Characterstics :- simple to implement, low bandwidth slow convergence.
- Examples :- Border Gateway Protocol, Routing information Protocol
- differences —

Characterstics	Shortest Path Routing	Path vector Routing
Knowledge of network Topology	complete map of the network.	only knows immediate neighbour and paths to other networks.
Algorithm used	Shortest path algorithm (eg. Dijkstra's)	Metric-based comparison of paths
convergence speed	Fast.	slow
scalability	High scalable	Moderately scalable
loop free	innately loop-free	Requires additional mechanism to prevent loops
complexity	more complex to implement	simpler to implement
Bandwidth	High	low
use cases	large, stable networks with high performance requirement.	small networks or network with frequent topology changes.

Q2. Differentiate betn. DHCP and ICMP.

Ans. DHCP : Dynamic Host Configuration Protocol (DHCP) is a network management protocol used to dynamically assign an IP address to any device, or node, on a network so they can communicate using IP. DHCP automates and centrally manages these configurations. There is no need to manually assign IP addresses to new devices. Therefore, there is no requirement for any user configuration to connect to a DHCP based network. DHCP can be implemented on local networks as well as large enterprise networks.

- ICMP : internet control Message Protocol (ICMP) is a network layer protocol used to diagnose communication errors by performing an error control mechanism. since IP does not have an inbuilt mechanism for sending error and control messages. it depends on internet control Message protocol (ICMP) to provide error control. ICMP is used for reporting errors and Management queries.

- Difference :-

DHCP	ICMP
• DHCP is used for dynamically assigning IP addresses and configuring network parameters automatically.	• ICMP is used for sending error message diagnostics, and checking network connectivity.
• DHCP operates at the app. layer (layer 7) of the OSI Model.	• ICMP operates at the network layer (layer 3) of the OSI model.
• DHCP involves the exchange of configuration information between a server and a client.	• ICMP does not carry user data but rather control message for network management.

Q3. Write a short note on hierachical Routing.

Ans. Hierachical routing is a network design approach that organizes ~~the~~ the global internet into a hierachical structure, making routing more scalable and efficient. It addresses the challenges associated with the growth of the internet by dividing the responsibility for routing and addressing into different levels or tiers.

In hierachical routing, routers are classified in groups called regions. Each router has information about the routers in its own region and it has no information about routers in other regions. So, routers save one record in their table for every other region.

Hierachical routing is a powerful and flexible routing strategy that can be used to improve the scalability, performance and manageability of large networks. However, it is important to consider the challenges of hierachical routing before deploying it in a network.

• Section – C (3×11 = 33)

Q1. Explain the Network layer designing issues.

Ans. Designing the network layer is a crucial aspect of creating a robust and efficient networking infrastructure. Several issues and considerations come into play when designing the network layer of a network. Here are some key network layer designing issues:

(i) Addressing scheme :-

• IPV4 vs IPV6 :- Choosing between IPV4 and IPV6 addressing schemes is a critical decision. IPV6 offers a large address space and improved features but may require significant infrastructure changes.

(ii) Routing Protocols :-

- Selection of Routing protocol: Choosing the appropriate routing protocol is essential. Factors such as network size, topology, and requirements influence the choice betn. protocols like RIP, OSPF, BGP.

(iii) Address Allocation :-

- Address Allocation Policies :- Defining policies for IP address allocation, including static vs. dynamic allocation and the use of technologies like DHCP.

(iv) ~~Sec~~ Security Considerations :-

- Access Control Lists : Implementing ACls for traffic filtering and controlling access to network resources.
- Virtual Private Networks (VPNs) :- integrating VPN technologies for secure comm. over public networks.

(v) Quality of Services (QoS) :-

- Traffic shaping and policing :- implementing mechanisms to control and shape network traffic to meet performance objectives.

(vi) Network Address Translation (NAT) :-

- Use of NAT :- Determining the need for Network Address Translation for conserving IP addresses and enhancing security.

(vii) Scalability - Designing the network layer with scalability in mind to accommodate future growth in terms of devices, users, and traffic volume.

(viii) Multicast Support :-

- Multicast Routing - Determining the need for multicast routing protocols to support efficient communication between multiple hosts.

Q2. Describe the concept of routing in the internet with diagram.

Ans: Routing is the process of selecting a path for data packets to travel from one network to another. When you send an email, your computer breaks it down into ~~sm~~ smaller packets of data. These packets are then sent to your internet service provider (ISP), which routes them to the destination email server.

The internet is a vast network of interconnected computers. In order for data packets to travel from one computer to another, they need to be routed through a series of networks. This is where routing tables come in. Routing tables are databases that contain information about the different networks on the internet and the paths that data packets can take to reach them.

- Concept of Routing in the internet :- Routing in the internet is the process of directing data packets between devices (routers) in a network to reach their intended destination. Routers are key components that make decisions about the best path for data to traval based on routing tables and protocols. The internet uses a hierachical routing approach, where routers are organized into layers to manage the large scale of the network.

- Routing Process :

(i) Source Device Sends data :- A device on the local network, like a computer, sends data to another device on the internet.

(ii) Local Router Decision :- The local router examines its routing table to determine the best path for the data.

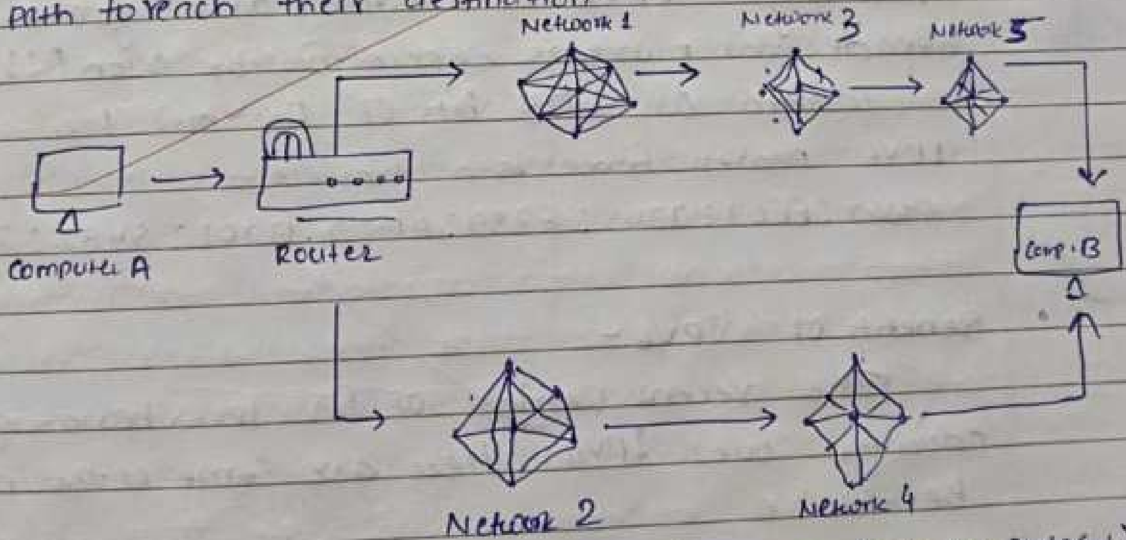
(iii) ISP Routing :- The data is passed to the ISP's router, which makes routing decisions based on its own routing table. The ISP's router is connected to other routers in the ISP network and the broader internet.

(iv) Destination ISP - The data eventually reaches the ISP connected to the destination network.

(v) Destination Router - The destination ISP's router makes decision based on it's routing table to direct the data toward the specific destination network.

(vi) Final destination :- The data arrives at the destination network and is delivered to the intended device.

This process involves multiple routers working together to ensure that data packets take the most efficient and reliable path to reach their destination.



Routing protocols, such as BGP (Border Gateway Protocol) and OSPF (Open Shortest Path First), play a crucial role in updating and maintaining the routing tables used by routers throughout the internet.

Q3. What is the difference betn. IPv4 and IPv6.

- IPv4 address consists of two things that are the network address and host address. It stands for internet protocol version 4. It was introduced in 1981 by DARPA. IPv4 addresses are 32-bit integers that have to be expressed in Decimal Notation. It is represented by 4 numbers seperated by dots in the range of 0-255, which have to be converted to 0 to 1, to be understood by computers. For ex - 189.123.123.90.

- IPv6 - IPv6 is based on IPv4 and stand for internet protocol version 6. it was first introduced in december 1995 by internet Engineering Task Force. IP version 6 is the new version of internet protocol, which is way better than IP version 4 in terms of complexity and efficiency. IPv6 is written as a group of 8 hexadecimal numbers seperated by colon (:). it can be written as 128 bits of 0s and 1s.
  IPv6 Address format ~~is a~~ example -
  ABCD:EF01:2345:6789:ABCD:B201:5482:D023

- Benefits of IPv6 -
  The recent version of IP IPv6 has a grater advantages over IPv4. Here are some of the mentioned benefits: -
- Larger Address space
- Improved security
- Simplified Header Format
- improved support for mobile Devices -

- Difference betn. IPv4 and IPv6 —

IPV4	IPV6
• IPv4 has a 32-bit address Length.	• IPv6 has a 128-bit address Length.
• it support manual and DHCP address configuration.	• it supports Auto and renumbering address configuration.
• in IPv4 end-to-end, connection integrity is unachievable.	• in IPv6 end-to-end, connection integrity is Achievable.
• it can generate 4.29×10^9 address space.	• The address space of IPV6 is quite large it can produce 3.4×10^{38} address space.
• Address representation of IPV4 is in decimal.	• address representation of IPV6 is in hexadecimal
• in IPv4 Checksum field is available.	• in IPv6 Checksum field is not available.
• in this, Encryption and authentication facility not provided.	• in IPv6, Encryption and authentication Facility are provided.
• it has a header of 20-60 bytes	• it has a header of 40 bytes fixed.
• Ex- of IPv4 - 66.94.29.13	• Ex. of IPV6 - 2001:0000:3238:DFE1:0063:0000:FEFB

The end